Financial Portfolio Management Based on Shaped-Based Unsupervised Machine Learning: A Dynamic Time Warping Baycenter Averaging Approach to International Markets and Periods of Downside Event Risks

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KEY FINDINGS

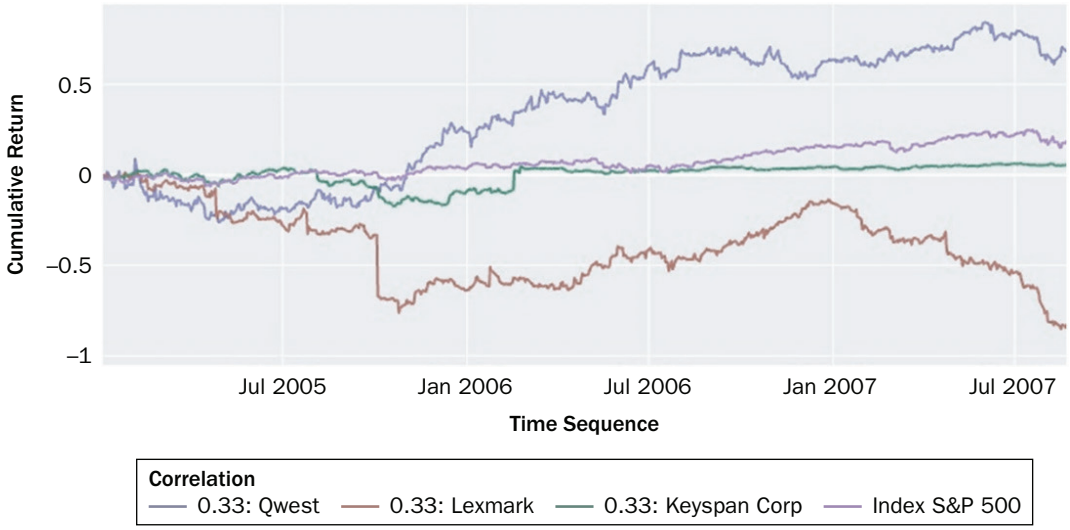
- Both back test and forward out-of-sample financial portfolio performance results showed consistent and significant outperformance in optimally weighted portfolios in selected international markets US, UK, China, and Australia, against correlation-based diversification methods.
- This outperformance persisted even across the 2008 global financial crisis, general market weakness in 2018, and the recent COVID-19 crisis, overcoming issues of correlation convergence and poor performance for correlation-based diversification methods.
- Persistency of financial portfolios were remarkably stable without significant persistency decay across a forward 10-year period, indicating strong industry value for portfolio selection, and predictive portfolio construction.

ABSTRACT

Empirical evidence has shown that modern portfolio theory relating to diversification had failed investors in the recent financial crises, times when investors would hope that diversification is an effective tool to sustain portfolio performance. Almost all markets around the world declined, with varying degrees, at the 2008 financial crisis and 2020 COVID-19 market crisis. Correlation-based diversification optimized portfolios were not spared, generating significant losses. Recent research on an unsupervised machine learning method of time-series clustering using Dynamic Time Warping (DTW) as a distance measure have shown research promise as a financial portfolio diversification method and shown prospects of overcoming correlation convergence issues during periods of downside event risks. This research validates the applicability of DTW cluster diversification to achieve persistent portfolio performance in international developed markets, even across periods of market weakness. Results showed outperformance of mean and median return and Sharpe metrics of optimally weighted DTW cluster diversification, against correlation-based diversification methods. The findings will augment existing literature in the use of data science approach to portfolio diversification.

EXHIBIT 1

Stocks at 0.33 Correlation to S&P 500 Index



Limitation with Use of Correlation in Modern Portfolio Theory

Using portfolio diversification to provide improved risk-return trade-off (Pourbabaee, Kwak, and Pirvu 2016) was already understood since the late seventeenth century (Sotiropoulos and Rutterford 2018; Carlos, Fletcher, and Neal 2015). However, it was only until the mathematical proofs of Markowitz (1952) of mean-variance optimization and diversification, when foundations were laid, showing how diversification can bring about reduction of portfolio risk for a given level of return (Theron and Vuuren 2018), through the selection of investment assets with minimal correlation in the portfolio (Medo, Yeung, and Zhang 2009).

The mathematical ingenuity of Markowitz (1952) formed the firm foundation of modern portfolio theory. However, one potential gap observed is the use of correlation in the selection of investment assets. F. J. Anscombe, in his landmark paper Graphs in Statistical Analysis (Anscombe 1973), described the observation of the classic Anscombe quartet, illustrating how different datasets can compute identical correlation, but exhibit markedly different data trends when plot on a graph. Based on data from the Singapore Stock Exchange (SGX) between 2015 to 2017, Lim and Ong (2021a) provided evidence of the markedly different observed price trends of securities, despite sharing the same correlation to the market index. We can also observe similar occurrences across other market indices, such as S&P 500 and FTSE 100. In Exhibits 1 and 2, we observe that although cumulative logarithmic returns are measured at 0.33 and 0.5 correlation to S&P 500 and FTSE 100 respectively, their return movements differ markedly.

This implies that relying on correlation to achieve portfolio diversification may not derive the most optimal multi-period portfolio risk-adjusted return, as securities in a portfolio can exhibit significantly different trends of cumulative logarithmic returns, even with the same computed pairwise correlation.

Empirical evidence have also shown that modern portfolio theory relating to diversification had failed investors in the recent financial crises (Theron and Vuuren 2018; Medo, Yeung, and Zhang 2009), times when investors would hope that diversification is an effective tool to sustain portfolio performance. Almost all markets around the world declined, with varying degrees, at the 2008 financial crisis (Price 2010). Mathematically computed mean-variance diversification optimized portfolios, that

EXHIBIT 2

Stocks at 0.5 Correlation to FTSE 100 Index

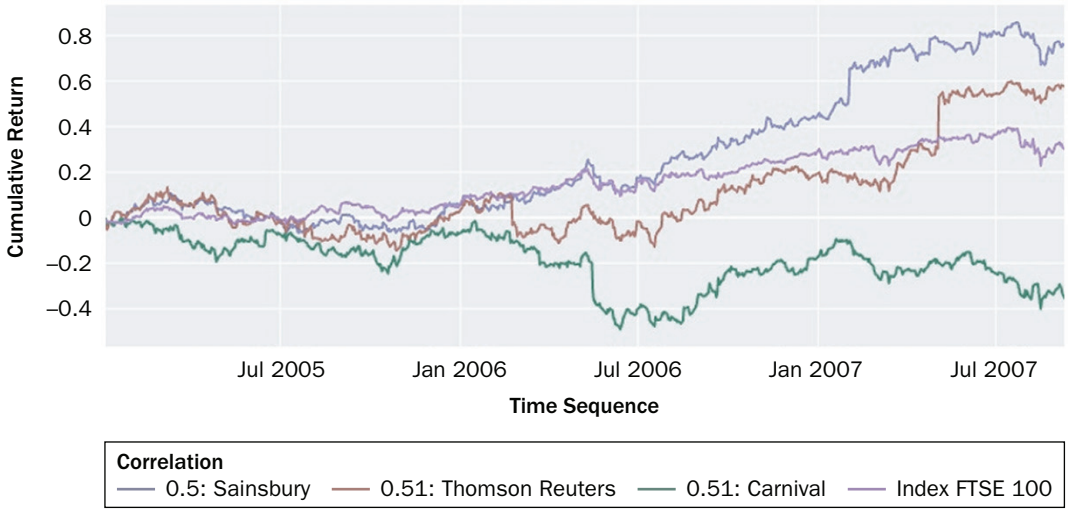
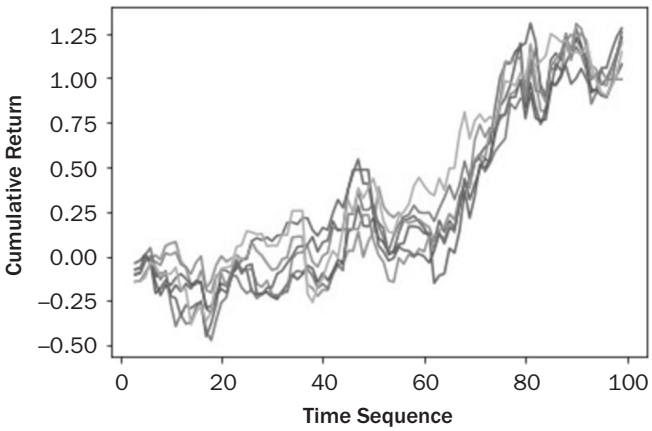


EXHIBIT 3

Illustration of Shape-Based Clustered Like-Trending Securities



used correlation as a model input, were not spared, generating significant losses (Harzallah and Abbes 2019; Theron and Vuuren 2018).

The reason why there are issues in terms of the use of correlation for time-series price data points, is because as a statistical measure to describe the strength of association between two time-series price data points, the relationship from the two variables is derived from the linear relationship between lines of best-fit drawn across the two scatter plot of time-series price data points. Linear relationships may not be the best descriptor of the trend of a set of dynamically moving time-series data points. Further, the static nature of linear approach requires model modifications that account for regime switching.

To put in simple terms, using correlation as a measure of time-series relationships between securities, is to use a blunt instrument to measure a precise measurement. To address the issues of correlation, it may be possible to explore alternative methods, such as the use of machine learning (ML), that can augment the theory (Lim and Ong 2021a, 2021b).

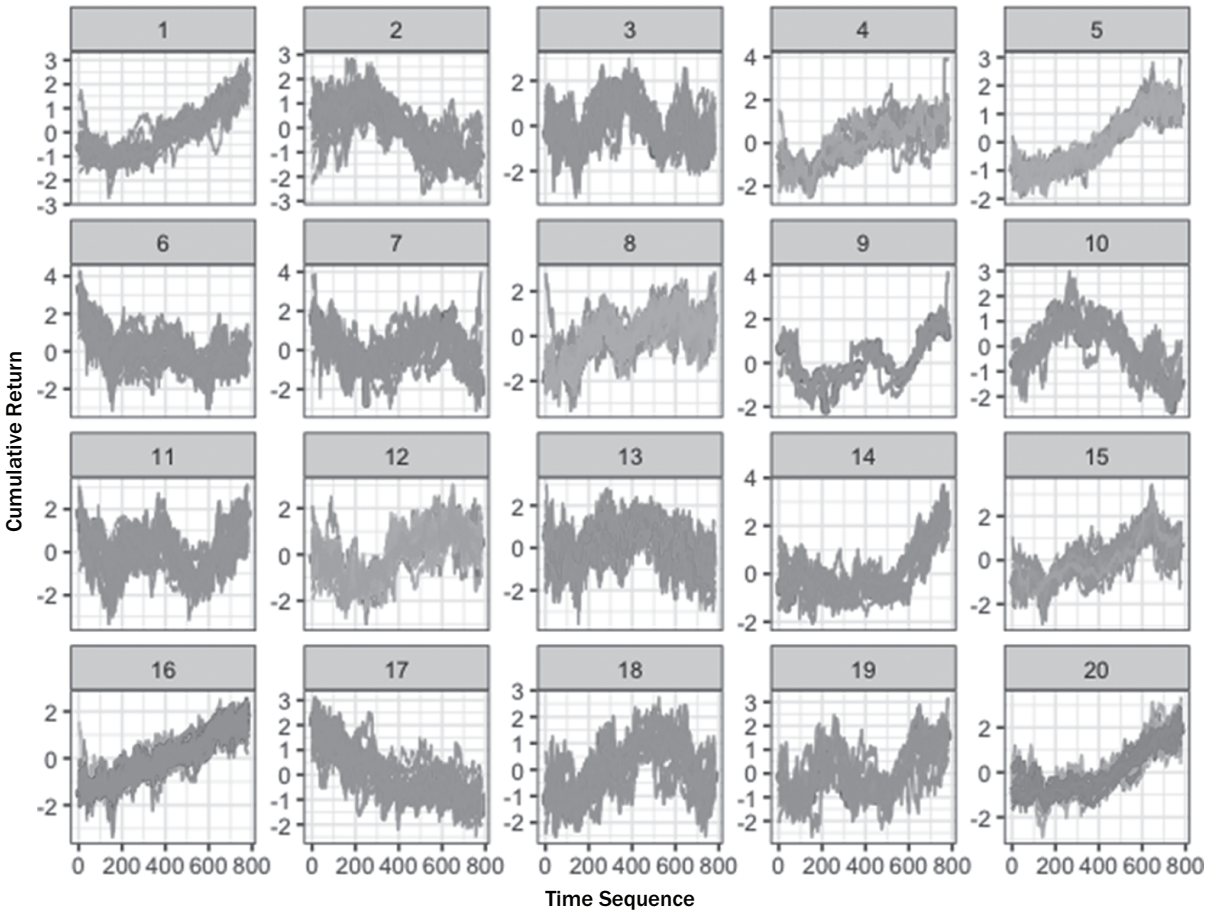
Proposed Method to Overcome Limitation of Modern Portfolio Theory

Exploratory analysis indicated a gap in current portfolio allocation and rebalancing methods and its objective of diversification. Intuitively, investors will achieve useful diversification if they invest in assets whose prices do not visually trend together across time.

Using shape-based clustering ML approach to identify clusters of assets that moves in similar (and dissimilar) trends over time, may bridge the gap and produce better portfolio performance. Investing in assets with dissimilar price trends, is akin to investing in uncorrelated assets with dissimilar dynamic time series data movements. Such shape-based clustering of assets that trend together over time when plotted in a time series graph can be implemented using an unsupervised clustering algorithm,

EXHIBIT 4

Illustration of 20-Cluster Shape-Based Clustered Like-Trending Securities



with Dynamic Time Warping (DTW) as a distance measure. In Exhibit 3, we observe an illustration of a group of like-trending securities that are clustered together based on the shape of their price trends. In Exhibit 4, we observe an illustration of the separation of a selected investment universe, into 20 distinct clusters of like-trending securities, based on the shape of their price trends.

This ML-driven method of clustering like-trending securities to find time-series relationship, is in sharp contrast to the correlation-based time-series relationship. Intuitively, it makes sense to perform portfolio allocation, by spreading out investments across different shape-based clusters, rather than concentrating investments within specific cluster(s). As demonstrated earlier, correlation is a blunt tool used to establish linear relationships, which is less precise vis-à-vis tools that account for dynamic price movements. Further, it is observed that similar correlation may generate markedly differing time series trends. This runs counter to the spirit to why correlation is used in portfolio selection in the first place—correlation is meant to act as an indicator for co-varying of time series trends. The use of new techniques to overcome weaknesses in existing methods is justifiable.

Research Contributions

Lim and Ong (2021a, 2021b) demonstrated robust time-persisted performance of agglomerative hierarchical clustering (AHC) algorithm, with DTW as a distance

EXHIBIT 5
Market Return Performance

Market	Index	2007	2008	2017	2018	2019	2020
Australia	ASX 100	12.0%	–41.7%	8.0%	–10.0%	24.9%	–1.9%
UK	FTSE 100	3.8%	–28.1%	8.0%	–11.7%	16.1%	–15.2%
China	CSI 300	161.6%	–66.0%	21.8%	–25.3%	36.1%	27.2%
US	S&P 500	3.5%	–33.5%	19.4%	–7.0%	30.3%	16.1%

measure, which has benefits in terms of: (i) portfolio allocation and rebalancing, (ii) dynamic predictive portfolio construction, and (iii) group or individual security selection through pattern recognition and outlier identification. However, the studies have two limitations:

1. Lim and Ong (2021a, 2021b) used equities exclusively from SGX. As such the applicability of the method to markets outside Singapore has not been tested.
   This study looks beyond the Singapore market, to major developed international markets. As a proxy, the following markets are selected, namely the: (i) Australian market represented by Australian market index ASX 100, which comprises Australia's large- and mid-cap securities, (ii) UK market represented by FTSE 100, which comprises UK's large- and mid-cap securities, (iii) Chinese market represented by CSI 300, which comprises China's large- and mid-cap securities, and (iv) US market represented by S&P 500, which comprises US large cap securities.
2. Input data in Lim and Ong (2021a, 2021b) contains only continuous daily price data from 2015 to 2018, a sideway trending period in the Singapore STI Index. It does not account specifically for performance in market downturns. This research aims to demonstrate if it is able to overcome the limitation of correlation convergence that plagues periods of market underperformance. We will be testing DTW-based cluster diversification over uptrend periods in: (i) 2007, (ii) 2017 and (iii) 2019. For out-of-sample testing, we will extend DTW-based cluster diversification to the respective following year of downside event risks and market weaknesses, specifically at the: (i) financial crisis at 2008, (ii) market weakness at 2018, and the (iii) recent 2019 COVID-19 crisis, when market performance in 2020 is overshadowed by COVID-19 concerns. A summary of the annualized market return performance is shown in Exhibit 5.

Further, a drawback of AHC-DTW at Lim and Ong (2021a, 2021b) is that it is computationally expensive and requires long runtime to execute. Petitjean, Ketterlin, and Gançarski (2011) and Petitjean et al. (2014) demonstrated outperformance, in terms of runtime efficiency, of centroid-based K-means clustering against hierarchical-based AHC, using an optimization method to average time series for DTW, known as DTW Baycenter Averaging. In this article, we utilize DTW Baycenter Averaging as the algorithm of choice.

As a proof-of-concept extension to Lim and Ong (2021a, 2021b), this work looks to validate the use of DTW-based cluster diversification across geographical markets and times of market weakness, with the aim of augmenting existing literature in portfolio diversification from a data science perspective. We hope to expand on our understanding of the methodology outlined in previous research by including a wider number of equities from different markets around the world and increasing the length of duration for the price data used for input. By doing so, we hope to test if

the proposed method is applicable for markets outside Singapore. In addition, during periods of downside event risks, where the correlations of equities increase greatly and the traditional notion of diversifying based on correlation is rendered less useful, we examine the applicability of DTW-based time series clustering.

LITERATURE REVIEW

Portfolio Selection and Diversification

Diversification of investments occurs when investors do not want to put all their eggs in one basket, and henceforth, seek to spread the risk of their portfolio across different investment assets. In this way, even as certain assets in the portfolio perform poorly, the overall portfolio would generally perform better over time. To best achieve this, Markowitz (1952) proposed to avoid investing in securities that tend to move together over time; investors should avoid adding assets to their portfolio when their prices exhibit high correlation. Sharpe ratio, a classic metric of relative investment performance, utilizes excess return for each percentage of risk borne by an investor to compare how attractive an investment is (Sharpe 1994). According to Markowitz (1952), the lower the correlation between portfolio assets, the higher the portfolio diversification, which in turn leads to the lowering of portfolio risk, and improvement to the Sharpe performance of a portfolio.

The most straightforward diversification technique is naïve diversification. Naïve diversification generates equally weighted portfolio that allows for a more diversified weight distribution because it gives the same allocation to all assets. However, this technique is naïve to the extent that it diversifies without accounting for security risk and return characteristics.

To achieve a desired portfolio performance, we can also model the portfolio selection problem by modelling investor preference based on an appropriate objective function. Portfolio optimization can be undertaken to compute an optimal distribution of assets and their weights, through the optimization of certain factors, such as risk or return, subjected to certain pre-specified constraints, such as limits placed on specific risk, industry, or geographical exposure.

For instance, mean variance optimization (MVO) of portfolios can be performed that generates the lowest risk for a given mean return. However, this technique suffers from extreme weights and inconsistent out-of-sample performance. A modification of the naive diversification technique is to allow for equal asset contribution to the risk of the portfolio, so that assets with lower volatility will receive more weights relative to highly volatile assets. This is known as the asset-risk parity approach (Maillard, Roncalli, and Teiletche 2010). However, this method does not work well when assets, while exhibiting different risks, are exposed to high correlation when there exist similar sources of risk and return (or factors). To overcome this, factor-risk parity (FRP) approach is proposed, which distributes asset risk across uncorrelated factors (Lassance, DeMiguel, and Vrins 2022). However, for high dimensional portfolios, the number of parameters can become intractable, which will require dimension reduction techniques, such as principal or independent component analysis, to reduce the uncorrelated factors to manageable dimensions. There also exists other diversification techniques that consider higher moments (e.g., kurtosis and skew), model return distributions (e.g., non-gaussian), and capture different risk measures (e.g., Value-at-Risk, semi-deviation) (Lassance and Vrins 2022).

Most portfolio diversification techniques, such as MVO and FRP, depend on inputs of observed and/or forecasted correlation of returns, as a mathematical measure of how asset returns trend together over time. However, as exposited earlier, there is room to improve existing correlation-based diversification techniques.

Portfolio Construction and Monitoring

Portfolio selection is tied to appropriate portfolio construction and monitoring practices. Portfolio allocation is the decision on how wealth of an institutional or individual investor can be distributed across assets. The process of asset allocation includes strategic asset allocation (SAA), which is the allocation of wealth to asset classes in a range of percentages (Brennan et al. 1997). SAA is said to be the key determinant of portfolio returns, accounting for 94% of the total variability of portfolio returns (Brinson et al. 1986). Other aspects of asset allocation include: (i) tactical asset allocation (TAA), which is the short-term deliberate deviation from SAA to take advantage of perceived investment opportunities in the market (Anson 2004), and (ii) individual security selection, which are attempts to generate higher returns than the asset class benchmark, by investing in individual assets that can generate higher relative expected returns (Staub 2006). Over time, investment portfolio typically undergoes drift, which is the process when, post-portfolio construction, the pre-specified asset allocation weights change as a result of changes in the value of the invested assets (Wermers 2012). Portfolio rebalancing is the process of monitoring and deciding how a portfolio, drifted from its initial asset allocation weights, shall be rebalanced to its initial weights or to new weights (Buetow et al. 2002). Asset allocation and rebalancing, security selection and short-term active deviation from the original portfolio mix to take advantage of trading opportunities, have significant influences in the total variability of portfolio returns, and they represent important decisions undertaken in a portfolio selection process, so as to achieve desired performance for the investor.

Machine Learning in Investing

The investment community is in its early stages of incorporating ML into its practices. According to a 2019 CFA Institute survey (CFA Institute 2019), relatively few investment professionals are currently using ML techniques in their investment processes. Most portfolio managers continue to rely on Excel and desktop market data tools; only 10% of portfolio manager respondents have used ML techniques in the past 12 months.

ML is a general term for computing methods and algorithms that allow machines to interpret inputs to uncover patterns to predict outputs (Rasekhschaffe and Jones 2019). ML techniques can augment human intelligence to enable investment professionals to reach a higher level of performance, enabling smarter decision making that leverages the collective intelligence of machines and humans.

ML programmes are popular with quantitative (systematic) managers who often find it helpful to apply these techniques to improve the effectiveness of their quantitative processes. Traditional statistics and econometrics are based on techniques first developed a couple of centuries ago, and their applications in finance often involve linear regression models. These linear models are effective in many situations. However, at least some of the complexities in the real world may be better captured using advanced ML techniques because of their ability to handle contextual and nonlinear relationships, which can often arise in finance, for example, in the handling of multicollinearity (Rasekhschaffe and Jones 2019). In these cases, ML techniques provide investment managers with additional toolkits that can give them an edge. Most ML research are applied on return forecasting, followed by portfolio construction in second place (Emerson et al. 2019).

There are different classes of ML techniques, including supervised learning, unsupervised learning (which includes cluster analysis), deep learning (DL), and reinforcement learning (RL).

Time-Series Clustering in Investing

As part of unsupervised learning, clustering is a data mining technique where related or homogeneous sets of similar data are grouped together without the prior knowledge of each group's members (Rai and Singh 2010).

There are various ways of clustering time-series (Fu 2011), namely shape-based, model-based and feature-based. In shape-based approach, the shapes of temporal sequenced data are as closely matched as possible, using conventional clustering methods such as K-means clustering or agglomerative hierarchical clustering (Vlachos, Lin, and Keogh 2003; Shumway 2003), with appropriate distance measures, such as dynamic time warping, Euclidean distance, or triangle similarity measure (Zhang et al. 2009; Faloutsos, Ranganathan, and Manolopoulos 1994; Sakoe and Chiba 1971), among others. In model-based approach, raw temporal sequenced data are transformed into model parameters, typically a parametric model for each temporal sequence, and clustering analysis is achieved through conventional clustering algorithms and appropriate distance measures (Warrenliao 2005). In feature-based approach, feature vectors, converted from raw temporal sequenced data, are clustered via conventional clustering algorithms and appropriate distance measures.

This study utilizes the shape-based clustering approach, as the latter two approaches suffer from limitations. Feature-based approach typically requires Euclidean distance measure as equal length feature vectors are computed from temporal data sequences (Hautamaki, Nykanen, and Franti 2008), whereas security prices may have differing temporal lengths due to the launch of companies' initial public offerings in the midst of the period of selection. Model-based approaches have limitations in terms of scalability issues (Vlachos, Gunopulos, and Das 2004), and its performance declines when clusters are close to one another (Mitsa 2009). Shaped-based time series clusters will normally reflect the fact that observations close together in time will be more closely related than observations further apart (Radhimeenakshi and Latha 2017). Shape-based clustering technique can be applied in several ways, including anomaly detection, correlation analysis, pattern recognition, prediction, and recommendation (Łuczak 2016).

Shaped-based clustering have been applied in financial markets in various contexts. Huang, Hsu, and Wang (2007) used a shape-based clustering algorithm known as extended visualization-induced self-organizing map algorithm to find and train the detection of repeatable temporal patterns and showed outperformance against momentum-based trading strategy (Narasimhan and Titman 1993). Dose and Cincotti (2005) applied shaped-based clustering in index tracking and found robust forecasting applications. Posch, Ullmann, and Wied (2019) found usefulness in the application of shaped-based clustering in identifying structural changes in financial markets or portfolios.

To apply clustering while accounting for time series movement of cumulative security returns, methods of similarity measures for time series data mining have been explored. The most used distance measure function is Euclidean distance. However, Euclidean approach has several shortcomings, including the need to compare datasets with the same time length, its inability to account for the negative effects of outliers or noise, and its high sensitivity to signal transformations (Iglesias and Wolfgang 2013). Introduced in the 1950s (Bellman and Kalaba 1959), DTW has been known to overcome these limitations with success. DTW allows comparison across different time lengths, many-to-one point comparison against one-to-one point comparison for Euclidean distance, and its results remain robust even with scaling and shifting (Cassisi et al. 2012).

DTW algorithm started in the early years in the field of speech recognition and image processing (Munich and Perona 1999; Rabiner and Juang 1993). Studies have

been conducted in various industries applying DTW with success (Keogh and Pazzani 2001), including the financial industry. Studies applying DTW in finance suggest that DTW can be valuable in the financial markets, which are highly dynamic. DTW is used to study patterns in the foreign exchange market (Wang et al. 2012), commodities market (Smiech 2015), and futures market (Kim et al. 2018), among others. DTW-based clustering had been effectively used in implementing trading strategies (Tsinaslanidis and Kugiumtzis 2014; Lee and Jeong 2012; Fleiss et al. 2020). Illustrating the profitability of the use of DTW-based clustering, research by Lee and Jeong (2012) combined DTW-based clustering with a basket of technical indicators, including Stochastic, MACD, and RSI indicators, and were able to achieve stable and profitable average of 10% return on KOSPI 200 index futures. In another example, Fleiss et al. (2020) achieved 6.57% excess return through the use of pattern-matching trading strategy on S&P 500 ETF in the pre-COVID-19 period, and 2.21% in post-COVID-19 period. Research on trading strategies involving DTW-based clustering demonstrated consistent profitability. Although studies have focused on performing pattern recognition and predictive security price trend movement, there is a lack of literature from the perspective of portfolio selection. Using DTW-based shape-based clustering in the context of portfolio selection, Lim and Ong (2021a, 2021b) found practical uses and significant improvement of portfolio performance.

DTW Baycenter Averaging

To perform shape-based clustering, this study utilizes DTW Baycenter Averaging (DBA) technique. DBA is a centroid-based clustering method that clusters time series based on their DTW distance. The novelty of the algorithm lies in the way that the centroid time series is constructed; constructing a measurement of centricity for time series is a complex task taking into account the DTW algorithm's ability to align the time series over time. The DBA method averages by first constructing a synthetic average time series (or medoid) using the barycenters between the time series points associated together in DTW distance calculation, and then iteratively refines the synthetic medoid time series by evaluating the associated time series used to construct it (Petitjean, Ketterlin, and Gançarski 2011).

Given a time series T with an ordered set of real values of length L, $T = \{t_1, \ldots, t_L\}$. A dataset D with a collection of N such time series is given by $D = \{T_1, \ldots, T_N\}$. To compute the average of a set of time series D in space E, the mean time series $\overline{T}$ is one that optimizes Equation 1, as follows:

$$\arg\min_{T \in E} \sum_{i=1}^{N} DTW^2(\overline{T}, T_i) \tag{1}$$

Using expectation-maximization optimization, DBA iteratively refines the mean time sequence $\overline{T}$, using the following:

1. Fixing $\overline{T}$, we find the most compact multiple alignment M of the set of sequences D with $\overline{T}$.
2. Fixing M, we update $\overline{T}$ as the best average sequence consistent with M.

Exhibit 6 provides the pseudocode of DBA (Petitjean et al. 2014). An implementation of DBA in Java and Matlab is provided in Petitjean (2014). DBA has been known to obtain lower residuals (Equation 1) via-a-vis other DTW-based clustering methods, making it one of the most competitive, if not the best, technique for time series averaging for DTW to date.

RESEARCH METHOD, DATA AND ASSUMPTIONS

Data and Analytical Tool

This research reviews international securities from ASX 100, FTSE 100, CSI 300 and S&P 500 market indices, as a proxy for developed markets beyond the Singapore market in Lim and Ong (2021a). As a proof-of-concept and to reduce computational runtime, for sampling purposes, the top 30% of securities by market capitalisation from these indices were selected. These top 30% of securities represent 72% of ASX 100, 76% of FTSE 100, 67% of CSI 300, 77% of S&P 500, by market capitalization.

The daily closing security prices were extracted from Bloomberg. Public holiday that fell on a weekday will result in blank data. These were imputed with the last closing price. Any equity with less than 250 daily closing price is excluded from the input dataset. The requirement of having at least 250 days of closing price data is to ensure that the stock has been traded for at least a trading year. Use of Bloomberg was consistent as a widely used data source for portfolio management investment practitioners.

EXHIBIT 6

DBA Algorithm for Time Series Clustering

Algorithm 6A: $DBA(D, I)$

Input: Set of sequence to average, D
Input: Number of iterations, I
Output: Best average sequence $\overline{T}$

```
//Compute medoid of set of sequences D
T̄ = medoid(D)
do I times T̄ = DBA_update(T̄, D)
return T̄
```

Algorithm 6B: $DBA_update(\overline{T_{init}}, D)$

Input: Average sequence of length L to refine, $\overline{T_{init}}$
Input: Set of sequence to average, D
Output: Best average sequence, $\overline{T}$

```
//Step 1: Compute multiple alignment for T̄_init
alignment = [∅, ..., ∅]   //L empty sets array
for each S in D do
    alignment_S = Multiple_alignment_DTW(T̄_init, S)
    for i = 1 to D do
      alignment[i] = alignment[i] ∪ alignment_S[i]
    done
done
//Step 2: Compute multiple alignment for the alignment
Let T̄ be a sequence of length L
for i = 1 to L do
    T̄(i) = mean(alignment[i])   //arithmetic mean of set
done
return T̄
```

(continued)

EXHIBIT 6 *(continued)*

DBA Algorithm for Time Series Clustering

Algorithm 6C: $DTW_multiple_alignment(S_{ref}, S)$

Input: Sequence to compute alignment, S_{ref}
Input: Sequence S to align to S_{ref} using DTW
Output: DTW alignment

```
//Step 1: Compute accumulated cost matrix of DTW
cost = DTW (S_ref, S)
//Step 2: Store all elements associated to S_ref
L = length(S_ref)
alignment = [∅, ..., ∅]   //L empty sets array
i = rows(cumulative_cost)   //i iterates over elements of S_ref
j = columns(cumulative_cost)   //j iterates over elements of S_ref
while i > 1 and j > 1 do
    alignment[i] = alignment[i] ∪ alignment_S[j]
    if i == 1 then j = j − 1
    else if j == 1 then i = i − 1
    else
        score = min(cost[i − 1][j − 1], cost[i][j − 1], cost[i − 1][j])
        if score == (cost[i − 1][j − 1] then
            i = i − 1
            j = j − 1
        else if score == cost[i − 1][j] then i = i − 1
        else j = j − 1
        end if
    end if
done
return alignment
```

Three one-year back-test periods were chosen, namely 2007, 2017, and 2019. These backtest periods preceded years of downside event risks experienced in 2008 sub-prime mortgage-induced crisis, 2018 general market weakness, and 2020 COVID-19-induced crisis. By applying DBA forward to 2008, 2018 and 2020, research will be able to ascertain if performance remains robust in out-of-sample years of financial downturns.

In total, the research sample comprises 298 securities in 2007–2008, 304 securities in 2017–2018, and 288 securities in 2019–2020.

Equations 2 and 3 were used to compute the geometric daily return (r_t) and annualized variance (σ_t^2) respectively, where d represents the number of trading days, and ω represents the weights of asset allocation. Portfolio weights sum to 1 and belong to a set of real numbers, $W := \{\omega \epsilon \mathbb{R} | 1'\omega = 1\}$. Covariance between the returns of asset i and j is represented by cov(σ) in Equation 4. Σ represents the computed covariance matrix. N represents the number of securities.

For a random variable X,

$$r_t \triangleq \ln R_t = \ln \frac{p_t}{p_{t-1}} \in \mathbb{R} \tag{2}$$

$$Var[X] \triangleq \sigma_t^2 = \omega_t' \, \Sigma \, \omega_t \in \mathbb{R}$$

$$\text{where } \Sigma = \begin{bmatrix} \sigma_{1,1} & \cdots & \sigma_{1,N} \\ \vdots & \ddots & \vdots \\ \sigma_{N,1} & \cdots & \sigma_{N,N} \end{bmatrix} \tag{3}$$

$$Cov[r_i, r_j] \triangleq \sigma_{ij} = \mathrm{E}[(r_i - \mathrm{E}[r_i])(r_j - \mathrm{E}[r_j])] \in \mathbb{R} \tag{4}$$

Portfolio return and Sharpe ratio were utilized to evaluate the performance of simulated portfolio results. According to Harvey and Liu (2015), it is a routine industry practice to discount the reported Sharpe ratios by 50%, among many other proposed haircut methods, in trading backtests. We reflect the 50% haircut in the relative Sharpe performance computation as shown in Equation 5. r_f represents the annualized risk-free rate. p represents the number of periods in the year.

$$\Delta S_{(t-1):t} \triangleq \frac{1}{2}\left(\frac{S_t}{S_{t-1}} - 1\right) \; \forall \; S_t = \left(\sqrt{p}\,\frac{r_t - r_f}{\sqrt{\mathrm{Var}[r_{(t-1):t}]}}\right) \in \mathbb{R} \tag{5}$$

Python was the main analytical tool and scripting language used for its simplicity and performance in the implementation of DBA. Python package *tslearn* was used.

The list of each equity's ISIN, GCIS sector, and daily closing price data were extracted from the Bloomberg Terminal using Microsoft Excel. For data crawling, we utilized VBA script to extract data using Excel Bloomberg terminal plugin (Exhibit 7).

Simulation of Portfolios

To study the performance of portfolios diversified based on DBA, the study simulated four portfolio test sets, namely: (i) mean variance optimization (or maximum Sharpe portfolio), (ii) minimum variance optimization, (iii) DBA, and (iv) correlation-based clustering.

Test sets (i) and (ii) had Sharpe maximization and risk minimization as objective functions. Portfolio selection of the asset constituents optimizes for optimal correlation equilibrium between the portfolio constituents in order to achieve these objective functions.

The latter test sets (iii) and (iv) applied clustering approaches; the former relied on DTW as a distance measure whereas the latter applied correlation as a dissimilarity measure. For portfolio selection of asset constituents, Sharpe maximization was applied, subjected to the constraint that asset selection would have to be made from different cluster groups. This constraint ensured that portfolio constituents were selected from dynamically dissimilar trending assets in the case of DBA, and dissimilar correlation groups in the case of correlation-based clustering. The objective of this test simulation was to study if DBA-augmented portfolio diversification outperforms correlation-based diversified portfolios, represented by mean variance optimization, minimum variance optimization and correlation-based clustering.

Test sets of unique 3-asset portfolios were generated. Portfolio size of 3-assets were selected for computational simplicity. The number of possible combinations (without repetitions) is computed as $H_k^n = \binom{k+n-1}{k}$, where n is the number of items we can select from, and k is the number of combinations (without repetitions) we want to obtain. Given the research sample comprises 298 securities in 2007–2008, 304

EXHIBIT 7

VBA Algorithm for Bloomberg Data Crawling

```
Sub saveRangeToCSV(saveRange As Range, csvName As String)
  Dim wb As Workbook
  Dim tempWB As Workbook

  Application.DisplayAlerts = False
  On Error GoTo err

  saveRange.Copy
  Set wb = ThisWorkbook

  Set tempWB = Application.Workbooks.Add(1)
  With tempWB
    .Sheets(1).Range("A1").PasteSpecial xlPasteValuesAndNumberFormats
    .SaveAs Filename:=csvName, FileFormat:=xlCSV, CreateBackup:=False
    .Close
  End With
err:
  Application.DisplayAlerts = True
End Sub
```

```
Sub GetData()

  Dim i As Integer
  Dim indexEnd As Integer, dataEnd As Integer
  Dim wb As Workbook
  Dim indexSheet As Worksheet, dataSheet As Worksheet
  Dim saveRange As Range
  Dim ticker As String, isin As String, exportFile As String
  Dim tickerArr As Variant

  Set wb = ThisWorkbook
  Set indexSheet = wb.Sheets(1)
  Set dataSheet = wb.Sheets("Data")

  indexEnd = indexSheet.Cells(indexSheet.Rows.Count, 1).End(xlUp).Row

  For i = 2 To indexEnd
    ticker = indexSheet.Range("A" & i).Value
    isin = indexSheet.Range("D" & i).Value

    dataSheet.Range("B1") = indexSheet.Range("A" & i)
    While dataSheet.Range("G1").Value <> 0:
      Application.Calculate
      Application.RTD.RefreshData
      DoEvents
    Wend

    dataEnd = dataSheet.Cells(dataSheet.Rows.Count, 1).End(xlUp).Row
    Set saveRange = dataSheet.Range("A7:F" & dataEnd)

    ticker = Replace(ticker, "Equity", isin)
    ticker = Replace(ticker, "/", "-")
    tickerArr = Split(ticker, " ")
    exportFile = wb.Path & "\" & Join(tickerArr, "_")
    Call saveRangeToCSV(saveRange, exportFile)
  Next i

End Sub
```

securities in 2017–2018, and 288 securities in 2019–2020, the number of possible combinations (without repetitions) for 3-asset portfolios are 4.46×10^6, 4.73×10^6 and 4.02×10^6 respectively. For computational simplicity and statistical robustness, we compute performance measures based on 1,000 unique portfolios.

The study simulated four portfolio test sets as follows:

- *Mean Variance Optimization (MeanVar)*: Also known as the *maximum Sharpe* portfolio, portfolio diversification was achieved through the maximization of Sharpe ratio approach, such that each simulated portfolios' combinations of asset pairwise correlation was the most optimal (or diversified) for Sharpe maximization (Zakamulin 2011).

  In this test set, 1,000 unique 3-asset portfolios were created, where each portfolio attained optimally diversified asset pairwise correlations such that Sharpe maximization was achieved. Refer to Equation 6 for optimization objective function:

$$\arg\max_{\omega \epsilon W} \frac{\omega_t' \mu}{\sqrt{\omega_t' \Sigma \omega_t}}, \text{subjected to } \omega_t' 1 = 1 \text{ and } \omega_i = \frac{1}{N}, \tag{6}$$

  where ω represents the weights of asset allocation, Σ represents the computed covariance matrix, *N* represents the number of securities, and μ represents mean return. When $\omega_i = \frac{1}{N}$, the objective function applies equal weightage to each portfolio constituent. We allow for short selling scenarios, when this constraint is relaxed.
- *Minimum Variance Optimization (MinVar)*: Portfolio diversification was achieved through the use of minimum variance optimization approach, such that each simulated portfolios' combinations of asset pairwise correlation was the most optimal (or diversified) for risk minimization (Choueifaty and Coignard 2008; Lhabitant 2017). MinVar is the classic diversification technique that improves Sharpe performance through risk minimization.

  In this test set, 1,000 unique 3-asset portfolios were created where each portfolio attained optimally diversified asset pairwise correlations such that risk minimization was achieved. Refer to Equation 7 for optimization objective function:

$$\arg\min_{\omega \epsilon W} \omega_t' \Sigma \omega_t, \text{subjected to } \omega_t' 1 = 1 \text{ and } \omega_i = \frac{1}{N}, \tag{7}$$

  where ω represents the weights of asset allocation, Σ represents the computed covariance matrix, and *N* represents the number of securities.
- *DBA (DTWClust)*: Portfolio diversification was achieved through the DBA. In this case, the lesser the DTW similarity achieved, the further apart their distances.

  In this test set, 1,000 unique 3-asset portfolios were created, each selected from different DBA cluster groups, such that diversification was achieved with optimal Sharpe performance. Refer to Equation 6 for Sharpe optimization objective function.

  To select the optimal cluster size, and hence the optimal *k* in clustering algorithms, research used the results of Dunn-Bonferroni statistical tests to identify statistically significant differences in the distributions of mean performance results, to reduce distribution overlaps. *k* is selected using the most optimal Dunn-Bonferroni statistical test results.

- *Correlation-based Clustering (CorrClust)*: This test set undertakes the same procedure as *DTWClust*, except correlation is used as the dissimilarity distance measure using complete linkage clustering. In this case, the less correlated the assets are, the further apart their distances. Complete linkage measures the distance using the farthest pair of observations in two clusters. This is a simple representation of clustering with the use of correlation as the key diversification metric (Raffinot 2017; León et al. 2017).

  In this test set, 1,000 unique 3-asset portfolios were created, each selected from different correlation cluster groups, such that diversification was achieved with optimal Sharpe performance. Refer to Equation 6 for Sharpe optimization objective function.

  Similarly, to select the optimal cluster size, and hence the optimal *k* in clustering algorithms, research used the results of Dunn-Bonferroni statistical tests to identify statistically significant differences in the distributions of mean performance results, to reduce distribution overlaps. *k* is selected using the most optimal Dunn-Bonferroni statistical test results.

Increased testing rigority was undertaken to further validate test results. Above four test sets were ran with optimization of weights, subjected to constraint of $\omega_t^T 1 = 1$, to allow for short selling scenarios. In total, eight test sets were ran. Analyses were performed to determine if results between simulated portfolios with equal weightage, were consistent with simulated portfolios with optimized weights.

Statistical Testing

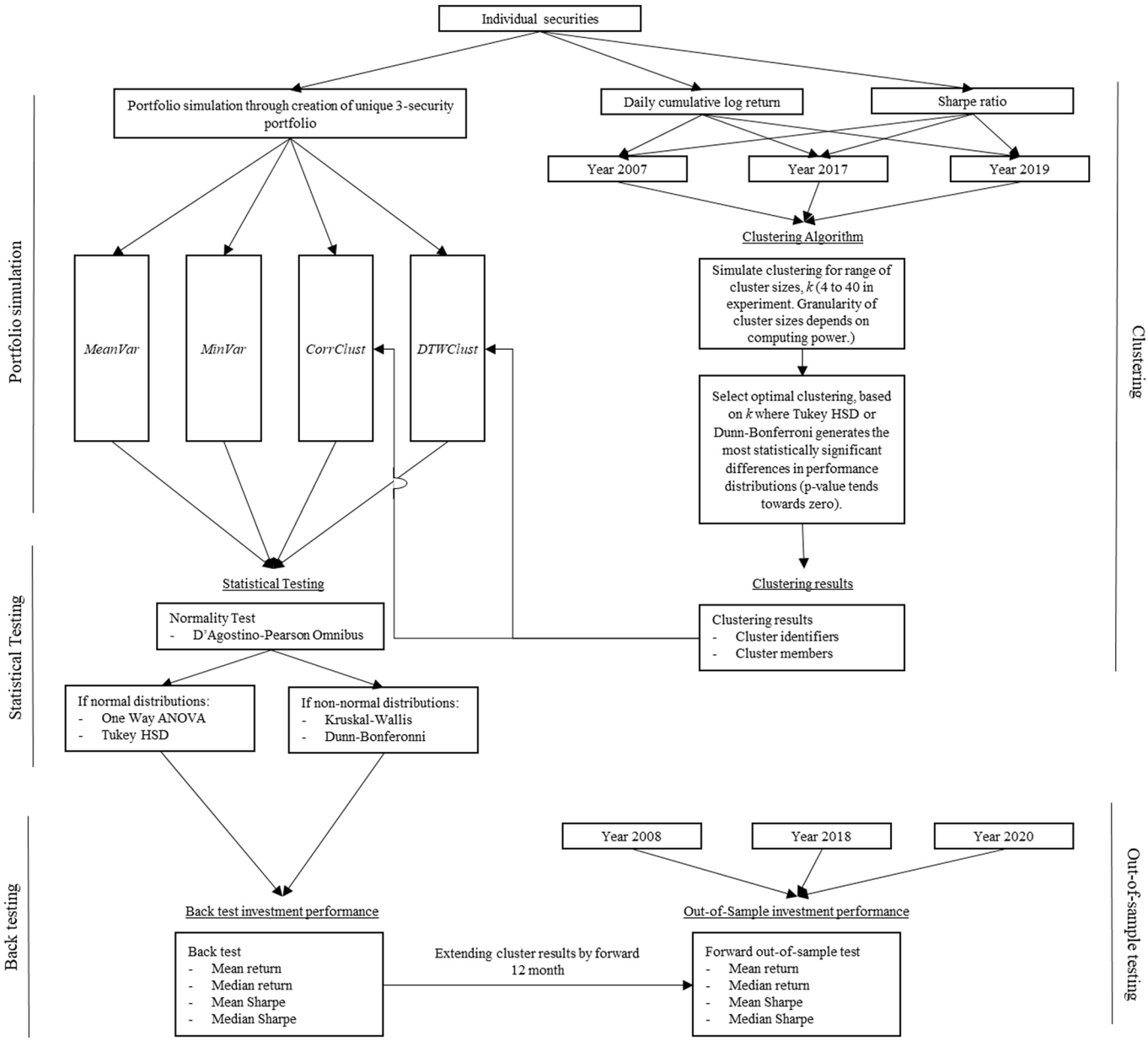
Post-simulation, mean, median, standard deviation and Sharpe ratio of the 1,000 securities in *MeanVar*, *MinVar*, *CorrClust*, and *DTWClust* portfolios were computed for the periods under observation and compared for performance, using both equal weightage and optimized weightage. Statistical tests were applied to determine how DTW-diversified portfolio performed, as compared to the other diversification methods and the control group.

To test the research objective, assessment of normality was conducted using D'Agostino-Pearson omnibus test (Öztuna, Elhan, and Tüccar 2006). This was followed by one-way ANOVA and Kruskal–Wallis tests. The former tests the statistical significance of the difference of the mean in two or more independent parametric distributions (Blanca Mena et al. 2017), and the latter is the non-parametric equivalent of one-way ANOVA. If normality tests determined all datasets to be of parametric distributions, research applied Tukey Honestly Significant Difference (HSD) all pairwise comparisons of group effects to test and identify meaningful differences in the 200 simulated portfolio mean returns and Sharpe ratios. Otherwise, research performed Dunn-Bonferroni test to identify statistically significant mean and median differences in performance results (Dinno 2015).

Summary of Methodology

Refer to Exhibit 8 for a summarized flowchart depicting the experimental methodology.

EXHIBIT 8
Methodology Flowchart



RESULTS AND DISCUSSION

Statistical Test Results

The performances of the portfolio sets were tested. Both mean return and Sharpe statistic of the four 1,000 portfolio displayed non-normal distributions. Applying the Kruskal-Wallis non-parametric test, the distributions showed that there were statistically significant differences in mean returns and Sharpe statistics between the different sets. The Dunn-Bonferroni ad-hoc test confirmed the difference between all test sets at 5% level of significance.

EXHIBIT 9

Backtest Results: 2007

12 Months Backtest (Equal Weight)	MeanVar	MinVar	CorrClust*	DTWClust*
Mean Return	13.42%	20.29%	2.35%	30.22%
Mean Standard Deviation	*22.39%*	*15.12%*	*5.83%*	*29.09%*
Median Return	11.52%	15.18%	–5.64%	29.25%
Median Standard Deviation	*19.83%*	*11.36%*	*22.47%*	*28.39%*
Mean Sharpe	0.51	1.21	0.06	0.97
Median Sharpe	0.48	1.16	–0.34	0.96
12 Months Backtest (Optimized Weight)	**MeanVar**	**MinVar**	**CorrClust****	**DTWClust****
Mean Return	60.13%	5.68%	152.26%	190.72%
Mean Standard Deviation	*30.43%*	*12.27%*	*40.28%*	*49.02%*
Median Return	49.47%	5.57%	129.15%	197.87%
Median Standard Deviation	*24.60%*	*9.39%*	*34.00%*	*51.01%*
Mean Sharpe	1.91	0.3	3.73	3.85
Median Sharpe	1.93	0.38	3.74	3.84

NOTES: *Optimal cluster number, k = 20. **Optimal cluster number, k = 25.

EXHIBIT 10

Backtest Results: 2017

12 Months Backtest (Equal Weight)	MeanVar	MinVar	CorrClust*	DTWClust*
Mean Return	19.11%	18.41%	34.73%	60.14%
Mean Standard Deviation	*11.72%*	*7.74%*	*12.44%*	*13.81%*
Median Return	18.94%	17.17%	43.45%	64.50%
Median Standard Deviation	*11.45%*	*7.19%*	*12.64%*	*14.57%*
Mean Sharpe	1.46	2.12	2.63	4.21
Median Sharpe	1.48	2.11	3.28	4.29
12 Months Backtest (Optimized Weight)	**MeanVar**	**MinVar**	**CorrClust****	**DTWClust****
Mean Return	35.53%	11.95%	67.00%	101.84%
Mean Standard Deviation	*15.17%*	*5.72%*	*14.77%*	*18.73%*
Median Return	32.84%	11.63%	53.12%	98.42%
Median Standard Deviation	*14.21%*	*5.87%*	*11.72%*	*18.76%*
Mean Sharpe	2.21	1.74	4.40	5.33
Median Sharpe	2.17	1.64	4.36	5.14

NOTES: *Optimal cluster number, k = 30. **Optimal cluster number, k = 35.

Backtest Results

From Exhibits 9 to 11, we observed outperformance of *DTWClust* portfolios against all other portfolios in both equally weighted and optimally weighted conditions.

Average Sharpe outperformance for all three periods, after accounting for 50% haircut for Sharpe ratio, ranged from 50.79% to 70.59% against *MeanVar* portfolios, 103.16% to 591.67% against *MinVar* portfolios, and 1.61% to 10.57% against *CorrClust* portfolios.

EXHIBIT 11

Backtest Results: 2019

12 Months Backtest (Equal Weight)	MeanVar	MinVar	CorrClust*	DTWClust*
Mean Return	23.60%	11.84%	29.84%	39.68%
Mean Standard Deviation	*15.77%*	*10.58%*	*16.28%*	*17.36%*
Median Return	23.96%	11.10%	27.46%	37.26%
Median Standard Deviation	*15.69%*	*10.11%*	*15.81%*	*18.56%*
Mean Sharpe	1.37	0.93	1.71	2.17
Median Sharpe	1.40	0.90	1.61	1.90
12 Months Backtest (Optimized Weight)	**MeanVar**	**MinVar**	**CorrClust****	**DTWClust****
Mean Return	38.25%	4.68%	57.44%	118.93%
Mean Standard Deviation	*18.98%*	*8.12%*	*15.49%*	*30.37%*
Median Return	34.60%	5.57%	50.51%	122.66%
Median Standard Deviation	*16.89%*	*8.93%*	*13.59%*	*31.50%*
Mean Sharpe	1.91	0.33	3.58	3.85
Median Sharpe	1.93	0.40	3.57	3.83

NOTES: *Optimal cluster number, k = 20. **Optimal cluster number, k = 25.

EXHIBIT 12

Out-of-Sample Test Results: 2008

12 Months Forward (Equal Weight)	MeanVar	MinVar	CorrClust*	DTWClust*
Mean Return	–67.85%	–26.35%	–67.60%	–129.91%
Mean Standard Deviation	*48.51%*	*27.00%*	*33.30%*	*48.14%*
Median Return	–61.86%	–22.13%	–89.18%	–135.02%
Median Standard Deviation	*45.94%*	*24.37%*	*46.05%*	*47.74%*
Mean Sharpe	–1.44	–1.05	–2.09	–2.74
Median Sharpe	–1.39	–0.99	–1.98	–2.87
12 Months Forward (Optimized Weight)	**MeanVar**	**MinVar**	**CorrClust****	**DTWClust****
Mean Return	–40.14%	–10.72%	–32.17%	41.20%
Mean Standard Deviation	*54.73%*	*48.92%*	*85.43%*	*156.80%*
Median Return	–45.91%	–6.59%	–12.78%	54.60%
Median Standard Deviation	*60.65%*	*19.09%*	*67.18%*	*114.35%*
Mean Sharpe	–0.77	–0.26	–0.40	0.25
Median Sharpe	–0.79	–0.45	–0.22	0.46

NOTES: *Optimal cluster number, k = 20. **Optimal cluster number, k = 40.

Out-of-Sample Test Results

To validate the results on the test dataset, further tests were conducted on a forward 12-month out-of-sample data—the 2008 global financial crisis, 2018 general market weakness, and 2020 COVID-19 crisis.

From Exhibits 12–14, for out-of-sample testing, it was noteworthy that equally weighted portfolio performed poorly, doing well for only the 2020 period. Weightage optimization appears to be necessary for outsized returns.

EXHIBIT 13

Out-of-Sample Test Results: 2018

12 Months Forward (Equal Weight)	MeanVar	MinVar	CorrClust*	DTWClust*
Mean Return	–13.94%	4.69%	–14.73%	–19.35%
Mean Standard Deviation	*18.53%*	*9.96%*	*20.16%*	*19.23%*
Median Return	–13.64%	5.22%	–9.82%	–15.87%
Median Standard Deviation	*17.98%*	*11.10%*	*19.38%*	*20.31%*
Mean Sharpe	–0.86	0.27	–0.83	–1.11
Median Sharpe	–0.87	0.29	–0.61	–0.88
12 Months Forward (Optimized Weight)	**MeanVar**	**MinVar**	**CorrClust****	**DTWClust****
Mean Return	–7.27%	7.04%	16.85%	28.43%
Mean Standard Deviation	*22.07%*	*10.96%*	*46.41%*	*44.05%*
Median Return	–7.86%	6.90%	3.43%	26.39%
Median Standard Deviation	*21.43%*	*10.21%*	*7.15%*	*38.11%*
Mean Sharpe	–0.42	0.46	0.32	0.60
Median Sharpe	–0.46	0.48	0.20	0.64

NOTES: *Optimal cluster number, k = 30. **Optimal cluster number, k = 35.

EXHIBIT 14

Out-of-Sample Test Results: 2020

12 Months Backtest (Equal Weight)	MeanVar	MinVar	CorrClust*	DTWClust*
Mean Return	1.96%	–7.59%	23.76%	29.82%
Mean Standard Deviation	*–0.80%*	*28.21%*	*29.01%*	*29.91%*
Median Return	3.68%	–6.53%	22.42%	27.60%
Median Standard Deviation	*28.00%*	*24.37%*	*26.52%*	*28.44%*
Mean Sharpe	0.05	–0.34	0.75	0.93
Median Sharpe	0.06	–0.35	0.77	0.90
12 Months Backtest (Optimized Weight)	**MeanVar**	**MinVar**	**CorrClust****	**DTWClust****
Mean Return	9.15%	26.38%	19.73%	48.50%
Mean Standard Deviation	*35.75%*	*64.16%*	*32.83%*	*54.71%*
Median Return	8.93%	28.47%	23.63%	39.25%
Median Standard Deviation	*38.50%*	*63.02%*	*39.33%*	*51.03%*
Mean Sharpe	0.2	0.38	0.54	0.85
Median Sharpe	0.18	0.42	0.55	0.73

NOTES: *Optimal cluster number, k = 20. **Optimal cluster number, k = 25.

When allowing for optimization of portfolio weightages, even after applying a 50% haircut to the return performances, the average return performances for *DTWClust* portfolios was better against *CorrClust* portfolios, ranging from 5.79% to 36.69%. Further, average *DTWClust* portfolios outperformed *MinVar* portfolios, ranging from 10.70% to 25.96%, and outperformed *MeanVar* portfolios, ranging from 17.85% to 40.67%.

On an overall basis, in the periods of market weakness for out-of-sample testing (Exhibit 5), results were robust, on both absolute and risk-adjusted returns. These results affirmed the persistency of *DTWClust* performance and the applicability of *DTWClust* across market downturns.

CONCLUSION

This research looks to determine the applicability of DTW-based cluster diversification in Lim and Ong (2021a, 2021b) beyond the Singapore market, in selected developed international markets, namely the Australian, UK, US and Chinese markets.

Further, research looks to determine the applicability of DTW cluster diversification across periods of market weakness, so as to validate the use of this diversification method, even during periods of correlation convergence at market downturns.

Both backtest and forward out-of-sample performance results showed consistent and significant outperformance in optimally weighted portfolios in international markets, even across the 2008 financial crisis, general market weakness in 2018, and the recent COVID-19 crisis, against correlation-based diversification methods. DBA-augmented portfolio diversification outperformed correlation-based diversified portfolios; the latter represented by mean variance optimization, minimum variance optimization and correlation-based clustering.

Persistency of *DTWClust* cluster members were remarkably stable and averaged around 50% across a forward 10-year period, without significant persistency decay. This explains the persistence of portfolio outperformance for the forward out-of-sample testing done in this research.

This research supports the use of DTW-based cluster diversification as a portfolio diversification method useful for portfolio allocation and rebalancing, and dynamic predictive portfolio construction. We recommend industry implementation of the solution, in terms of the creation of recommender systems and integration into investment decision-making processes.

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